

# TTC Domains<sup>\*</sup>

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## Abstract

For the object reallocation problem, we study whether characterizations of Top Trading Cycles (TTC) based on individual rationality, efficiency, and strategyproofness on the unrestricted domain extend to restricted preference domains. We introduce the top-two condition and show that it offers a useful criterion for answering this question. The condition requires that, within every subset of objects, any two objects that can each be ranked first can also be ranked as the top two, in both possible orders. We first show that this condition is sufficient: on every domain satisfying the top-two condition, TTC is the unique rule satisfying the relevant axioms. We also provide a partial converse. For domains that fail the top-two condition within a small subset of objects and satisfy a mild extension condition, we construct a rule distinct from TTC satisfying these axioms. Our results provide a unifying perspective on existing findings for specific domains, such as the single-peaked and single-dipped domains, while also addressing several previously unexplored domains, including the circular and partial-agreement domains.

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# 1 Introduction

This paper studies the object reallocation problem, introduced by Shapley and Scarf (1974). Each agent is endowed with an indivisible object and has a strict preference over all objects. A rule maps reported preferences into a reallocation of the objects.

A fundamental rule for this problem is Top Trading Cycles (TTC). Shapley and Scarf (1974) proposed TTC, credited to David Gale, as an algorithm for finding an allocation in the core, and Roth and Postlewaite (1977) later proved that it produces the unique core allocation. Focusing on incentives, Roth (1982) showed that TTC is strategyproof. In a seminal result, Ma (1994) characterized TTC as the unique rule satisfying individual rationality, Pareto efficiency, and strategyproofness. Alternative proofs were later provided by Svensson (1999), Anno (2015), Sethuraman (2016), Bade (2019). More recently, Ekici (2024) strengthened this characterization by replacing Pareto efficiency with pair efficiency, a substantially weaker requirement that only rules out mutually beneficial swaps between two agents. Ekici and Sethuraman (2024) provide a short proof of this result. Alternative characterizations using axioms such as group strategyproofness (Bird (1984), Takamiya (2001)), independence of irrelevant rankings (Morrill (2013)), non-bossiness (Miyagawa (2002), Ehlers (2014)), and endowments-swapping-proofness (Fujinaka and Wakayama (2018)) have been established.<sup>1</sup>

Despite these strong properties, there are preference profiles at which the TTC allocation may be less appealing than alternative allocations. To illustrate, consider the profile in Table 1. Under TTC, only agents 1 and 2 are assigned their first-ranked objects, while all other agents receive their second-ranked objects. However, if agent 1 is instead assigned their second-ranked object, every other agent can receive their first-ranked object. Thus, TTC may preclude the execution of arbitrarily long trading cycles. This raises the question of whether such alternative allocations can be selected while retaining strategyproofness. One possible avenue is to exploit features of the environment that make certain preference rankings implausible, thereby restricting the preference domain. By limiting agents' opportunities to misrepresent their preferences, such restrictions weaken the force of strategyproofness and potentially enrich the class of rules satisfying the axioms.

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<sup>1</sup>Some related strands of literature focus on reallocation problems tailored to specific environments (Abdulkadiroğlu and Sönmez (1999), Roth, Sönmez, and Ünver (2004), Schummer and Vohra (2013)) and object allocation problems (Carroll (2014), Hylland and Zeckhauser (1979), Pápai (2000), Pycia and Ünver (2017)). Morrill and Roth (2024) survey this literature and highlight the relevance of TTC in these environments.

| $P_1$    | $P_2$    | $P_3$    | $\cdots$ | $P_k$     | $\cdots$ | $P_{n-1}$ | $P_n$     |
|----------|----------|----------|----------|-----------|----------|-----------|-----------|
| $o_2$    | $o_1$    | $o_2$    | $\cdots$ | $o_{k-1}$ | $\cdots$ | $o_{n-2}$ | $o_{n-1}$ |
| $o_n$    | $o_2$    | $o_3$    | $\cdots$ | $o_k$     | $\cdots$ | $o_{n-1}$ | $o_n$     |
| $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ | $\vdots$  | $\ddots$ | $\vdots$  | $\vdots$  |

Table 1: A preference profile in which TTC precludes the execution of a longer trading cycle.

This motivates our central question: under what domain restrictions do the characterizations of TTC by Ma (1994) and Ekici (2024) continue to hold? We address this question by introducing a richness condition on domains, which we call the *top-two condition*. The condition requires that, within every subset of objects, if two objects can each be ranked first, then they can also be ranked as the top two objects in both possible orders.

We first show that the top-two condition is sufficient: if a domain satisfies the top-two condition, then TTC is the only rule that is individually rational, pair efficient, and strategyproof. To illustrate the idea, consider again the profile in Table 1. If the domain allows agent 1 to report a preference that keeps their first-ranked object unchanged while moving their endowment up to second rank, then individual rationality and pair efficiency imply that the rule must assign agent 1 their first-ranked object. Strategyproofness then implies that the rule must do so at the original profile in Table 1 as well, as is the case under TTC. More generally, at any profile, the condition ensures that agents in a top trading cycle can move their endowments just below their first-ranked objects. Such auxiliary profiles are a key ingredient in existing proofs of the characterizations of Ma (1994) and Ekici (2024). At these profiles, while Pareto efficiency directly implies that trading cycles must be executed, Ekici (2024) and Ekici and Sethuraman (2024) exploit the richness of the unrestricted domain to establish the same conclusion under pair efficiency. In comparison, we show that this conclusion follows with no additional richness beyond the top-two condition, thereby identifying a broad class of domains on which both characterizations continue to hold.

Turning to necessity, although we do not establish it in full generality, we show that for domains that fail the top-two condition in a certain way, the axioms do not pin down TTC. Specifically, if there exists a small subset of objects within which the failure occurs and every object outside this subset can be ranked first relative to the objects in the subset, then we construct a non-TTC rule that is individually rational, Pareto efficient, and strat-

egyproof. We first construct a baseline rule for small economies. This rule deviates from TTC at profiles of the form displayed in Table 1: assuming agent 1 cannot rank their endowment second, the rule assigns agent 1 their second-ranked object instead of the first-ranked object they would receive under TTC. By carefully selecting the profiles at which this deviation occurs, we ensure that the resulting rule satisfies the axioms. We then use this baseline construction to define, for economies of arbitrary size, a non-TTC rule satisfying the axioms.

Together, our results provide a unifying perspective on existing findings for specific domains while also addressing several previously unexplored domains. By the sufficiency of the top-two condition, the single-dipped domain, previously studied by Tamura (2023), Hu and Zhang (2024), and the partial-agreement domain, which contains all preferences consistent with a predefined partial order, are TTC domains. Our necessity result covers the single-peaked domain, previously studied by Bade (2019), Tamura and Hosseini (2022), and the circular domain, previously studied in social choice by Sato (2010), implying that neither is a TTC domain. Although our results fall short of a full characterization of TTC domains, they establish the top-two condition as a valuable tool for analyzing object reallocation problems under domain restrictions.<sup>2</sup>

## 2 Model

### Preliminaries

Let  $N = \{1, \dots, n\}$  be a finite set of agents. Let  $O = \{o_1, \dots, o_n\}$  be a finite set of objects such that  $o_i$  denotes agent  $i$ 's endowment. Agents have strict preferences over objects. We denote by  $\mathcal{P}$  the set of all strict linear orders over  $O$ , and we let  $\mathcal{D} \subset \mathcal{P}$  denote a preference domain. Let  $P = (P_i)_{i \in N} \in \mathcal{D}^N$  denote a preference profile where  $P_i \in \mathcal{D}$  is agent  $i$ 's preference. Following standard convention, for  $S \subset N$ , we let  $P_S = (P_i)_{i \in S}$  and  $P_{-S} = (P_i)_{i \in N \setminus S}$ . For each  $P_0 \in \mathcal{D}$ , we denote by  $R_0$  the “at least as desirable as” relation associated with  $P_0$ , i.e., for each pair  $o, o' \in O$ ,  $o R_0 o'$  if and only if either  $o P_0 o'$  or  $o = o'$ . We refer to a set of agents, their endowments, and their preferences over these objects as an *economy*.

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<sup>2</sup>In a similar spirit, Alcalde and Barbera (1994) propose the top-dominance criterion for two-sided matching, and Aswal, Chatterji, and Sen (2003) introduce the notion of linked domains in social choice. In early work, Sen and Pattanaik (1969) identify domain restrictions ensuring transitivity of the majority relation. A survey by Elkind, Lackner, and Peters (2022) reviews a wide range of domain restrictions, some of which may be relevant in our context.

An *allocation*  $x : N \rightarrow O$  is a bijection that assigns to each agent an object. Let  $\mathcal{X}$  be the set of allocations. For each  $x \in \mathcal{X}$  and each  $i \in N$ , we denote by  $x_i \in O$  the assignment of agent  $i$  at  $x$ . A *rule*  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  associates with each preference profile  $P \in \mathcal{D}^N$  an allocation.

## Axioms

We now introduce some standard properties of allocations and rules.

Given  $P \in \mathcal{D}^N$ , an allocation  $x \in \mathcal{X}$  is *individually rational* at  $P$  if for each  $i \in N$ ,  $x_i R_i o_i$ . A rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  is *individually rational* if for each  $P \in \mathcal{D}^N$ ,  $\varphi(P)$  is individually rational at  $P$ .

Given  $P \in \mathcal{D}^N$ , an allocation  $x \in \mathcal{X}$  is *Pareto efficient* at  $P$  if there is no allocation  $y \in \mathcal{X}$  such that for each  $i \in N$ ,  $y_i R_i x_i$  and for some  $j \in N$ ,  $y_j P_j x_j$ . A rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  is *Pareto efficient* if for each  $P \in \mathcal{D}^N$ ,  $\varphi(P)$  is Pareto efficient at  $P$ . Given  $P \in \mathcal{D}^N$ , an allocation  $x \in \mathcal{X}$  is *pair efficient* at  $P$  if there is no  $i, j \in N$  such that  $x_j P_i x_i$  and  $x_i P_j x_j$ . A rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  is *pair efficient* if for each  $P \in \mathcal{D}^N$ ,  $\varphi(P)$  is pair efficient at  $P$ .

A rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  is *strategyproof* if for each  $P \in \mathcal{D}^N$ , there is no  $i \in N$  and  $P'_i \in \mathcal{D}$  such that  $\varphi_i(P'_i, P_{-i}) P_i \varphi_i(P)$ . A rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  is *group strategyproof* if for each  $P \in \mathcal{D}^N$ , there is no nonempty  $S \subset N$  and  $P'_S \in \mathcal{D}^S$  such that for each  $i \in S$ ,  $\varphi_i(P'_S, P_{-S}) R_i \varphi_i(P)$  and for some  $j \in S$ ,  $\varphi_j(P'_S, P_{-S}) P_j \varphi_j(P)$ .

## Top Trading Cycles

We now describe the TTC rule. For any profile  $P \in \mathcal{P}^N$ , the TTC algorithm determines an allocation as follows:

- (i) Each agent points to the agent who owns their most-preferred object.
- (ii) In the resulting directed graph, there is at least one cycle. All agents belonging to cycles are assigned the objects they point to and leave the economy.
- (iii) If agents remain in the economy, repeat the procedure with the remaining agents and objects. Otherwise, terminate.

We let  $TTC(P) \in \mathcal{X}$  denote the allocation that results from running this algorithm at profile  $P \in \mathcal{P}^N$ . For any domain  $\mathcal{D} \subset \mathcal{P}$ , the *TTC rule* is the rule that assigns to each  $P \in \mathcal{D}^N$  the allocation  $TTC(P)$ . It is well known that TTC satisfies all the properties defined above on the unrestricted domain. Hence, the same properties hold on every domain  $\mathcal{D} \subset \mathcal{P}$ .

**Fact 1.** *For any  $\mathcal{D} \subset \mathcal{P}$ , the TTC rule is individually rational, Pareto efficient, and group strategyproof.*

For the unrestricted domain  $\mathcal{D} = \mathcal{P}$ , there is, in fact, no other rule that satisfies even the weak combination of individual rationality, pair efficiency, and strategyproofness (Ekici (2024)). For brevity, we call such domains *TTC domains*.<sup>3</sup> Our goal is to provide a general characterization of these domains.

## Notation

We will sometimes describe a preference  $P_0 \in \mathcal{D}$  by listing the objects in  $O$  in the order specified by  $P_0$ . For example,  $P_0 \in \mathcal{D}$  such that  $o_k P_0 o_{k+1}$  for all  $k$  can be succinctly represented as  $o_1 o_2 \dots o_n$ .

For any  $P_0 \in \mathcal{D}$  and  $O' \subset O$ , we let  $r_k(P_0, O')$  denote the object ranked  $k$ -th under  $P_0$  restricted to  $O'$ . For example,  $r_1(P_0, O')$  is the object ranked first according to  $P_0$  among  $O'$ .

Similarly, for any  $\mathcal{D}' \subset \mathcal{D}$  and  $O' \subset O$ , we let  $r_k(\mathcal{D}', O')$  denote the set of objects that are ranked  $k$ -th according to some  $P_0 \in \mathcal{D}'$  restricted to  $O'$ . Formally,

$$r_k(\mathcal{D}', O') = \{o' \in O' : \text{there exists } P_0 \in \mathcal{D}' \text{ such that } r_k(P_0, O') = o'\}.$$

For example,  $r_1(\mathcal{D}, O')$  is the set of objects ranked first according to preferences in  $\mathcal{D}$  among  $O'$ .

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<sup>3</sup>The terminology follows a common convention in the literature on domain restrictions, where a family of domains is identified by the class of rules it characterizes; see, for example, Aswal, Chatterji, and Sen (2003) on dictatorial domains.

## 3 Results

### 3.1 Top-two condition

We now introduce our richness condition on preference domains. We first define the condition within a fixed subset of objects. A domain satisfies the top-two condition within  $O' \subset O$  if any two objects that can be ranked first within  $O'$  can also be ranked as the top two within  $O'$ , in both possible orders.

**Definition 1.** A domain  $\mathcal{D} \subset \mathcal{P}$  satisfies the *top-two condition within*  $O' \subset O$  if for any distinct  $a, b \in r_1(\mathcal{D}, O')$ , there exists  $P_0 \in \mathcal{D}$  such that

$$a = r_1(P_0, O') \quad \text{and} \quad b = r_2(P_0, O').$$

**Definition 2.** A domain  $\mathcal{D} \subset \mathcal{P}$  satisfies the *top-two condition* if, for each  $O' \subset O$ ,  $\mathcal{D}$  satisfies the top-two condition within  $O'$ .

We now illustrate the condition with some examples. For any domain  $\mathcal{D} \subset \mathcal{P}$ , the top-two condition within  $O'$  is trivially satisfied whenever  $|O'| \leq 2$ , so it suffices to check subsets of size at least three. Consider

$$\mathcal{D}_1 = \{o_1 o_2 o_3, o_2 o_3 o_1, o_2 o_1 o_3\}$$

and

$$\mathcal{D}_2 = \{o_1 o_2 o_3, o_2 o_3 o_1, o_1 o_3 o_2\}.$$

In both domains, only  $o_1$  and  $o_2$  can be ranked first within  $O$ . In  $\mathcal{D}_1$ , these two objects can be ranked as the top two in both possible orders, so  $\mathcal{D}_1$  satisfies the top-two condition. In  $\mathcal{D}_2$ , however, no preference ranks  $o_2$  first and  $o_1$  second, so  $\mathcal{D}_2$  does not satisfy the top-two condition. The condition can also fail on a strict subset of objects. Consider

$$\mathcal{D}_3 = \{o_1 o_2 o_3 o_4, o_1 o_3 o_2 o_4, o_2 o_1 o_4 o_3, o_2 o_4 o_3 o_1\}.$$

Although  $\mathcal{D}_3$  satisfies the top-two condition within  $O$ , it fails the condition within  $O' = \{o_1, o_3, o_4\}$ . Within this subset, both  $o_1$  and  $o_4$  can be ranked first, but no preference ranks  $o_4$  first and  $o_1$  second. Thus,  $\mathcal{D}_3$  fails the top-two condition.

The top-two condition is reminiscent of some existing richness conditions in the literature. One is *free pair at the top (FPT)*, which requires that any pair of objects  $a, b \in O$  can be

ranked as the top two (Chatterji and Sen (2011)). Another is Assumption B of Sönmez (1999), which, adapted to our setting, requires that if a preference ranks  $a$  above  $b$ , then it must be possible to move  $b$  immediately below  $a$ , while leaving unchanged the sets of objects ranked above and below  $a$ .<sup>4</sup> Both conditions are stronger than the top-two condition: any domain satisfying FPT or Assumption B satisfies the top-two condition, but the converse does not hold. For example, when  $n \geq 3$ , the singleton domain  $\mathcal{D} = \{o_1 o_2 \dots o_n\}$  satisfies the top-two condition, but satisfies neither FPT nor Assumption B.

### 3.2 Top-two is sufficient

The following result shows that if a domain satisfies the top-two condition, it is a TTC domain.

**Theorem 1.** *Suppose  $\mathcal{D} \subset \mathcal{P}$  satisfies the top-two condition. Let  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  be a rule. The following are equivalent:*

- (i)  $\varphi$  is the TTC rule.
- (ii)  $\varphi$  is individually rational, Pareto efficient, and strategyproof.
- (iii)  $\varphi$  is individually rational, pair efficient, and strategyproof.

To prove this result, we show that if  $\varphi$  satisfies (iii), then at any profile  $P \in \mathcal{D}^N$ , it must coincide with TTC. Consider first the agents involved in a trading cycle in the first round of TTC. Since  $\mathcal{D}$  satisfies the top-two condition, these agents can report their endowments as their second-ranked objects. At this auxiliary profile, individual rationality implies that  $\varphi$  must either execute the cycle or assign all these agents their endowments.

While Pareto efficiency would immediately imply that  $\varphi$  must execute the cycle, the core of our argument is to show that the same conclusion follows under the weaker requirement of pair efficiency (Lemma 1). Suppose the cycle is  $S = \{i_1, \dots, i_{|S|}\}$ , as in Table 2. Focus on agent  $i_1$ . We show that  $\varphi$  must assign agent  $i_1$  object  $o_{i_k}$  if they instead report a preference  $P_{i_1}^k \in \mathcal{D}$  that ranks  $o_{i_k}$  first and  $o_{i_{k+1}}$  second, as in Table 3. The base case  $k = |S|$  follows

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<sup>4</sup>Sönmez (1999) studies a more general object reallocation problem, in which agents may own multiple objects and have a preference over allocations, rather than only over their own assignments. Under a no-indifference condition and Assumption B on the preference domain, the paper establishes the significance of the core correspondence being (essentially) single-valued for existence of rules satisfying individual rationality, Pareto efficiency, and strategyproofness.



directly from individual rationality and pair efficiency. For the induction step, assuming the claim holds for  $k + 1$ , strategyproofness implies that when  $i_1$  reports  $P_{i_1}^k$ ,  $\varphi$  must assign  $i_1$  either  $o_{i_k}$  or  $o_{i_{k+1}}$ . Pair efficiency rules out  $o_{i_{k+1}}$ , so  $i_1$  must receive  $o_{i_k}$ . In particular,  $\varphi$  assigns  $i_1$  object  $o_{i_2}$  when they report  $P_{i_1}^2$ . By strategyproofness,  $\varphi$  must assign  $o_{i_2}$  to  $i_1$  at the auxiliary profile, and individual rationality then implies that  $\varphi$  must execute the cycle  $S$ .

Finally, strategyproofness and individual rationality imply that  $\varphi$  must execute the cycle at the original profile  $P$  as well. Since the argument does not depend on the reports of agents outside the cycle, we can remove these agents and their assigned objects and apply the same argument to the agents in the next TTC cycle, as  $\mathcal{D}$  satisfies the top-two condition within the reduced object set. Iterating over subsequent cycles yields  $\varphi(P) = TTC(P)$ . We now give the formal proof.

*Proof.* The implications  $(i) \implies (ii) \implies (iii)$  are immediate, so it suffices to prove  $(iii) \implies (i)$ .

Let  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  be individually rational, pair efficient, and strategyproof. Fix  $P \in \mathcal{D}^N$  and let  $x = TTC(P)$ . We show that  $\varphi(P) = x$ .

We first show that  $\varphi$  executes the cycle in the first round of TTC at  $P$ . Let  $S \subset N$  be the set of agents in this cycle. If  $|S| = 1$ , individual rationality immediately gives  $\varphi_i(P) = x_i$  for the unique  $i \in S$ . Hence assume  $|S| \geq 2$ . For any  $i \in S$ , both  $x_i, o_i \in r_1(\mathcal{D}, O)$ , and since  $\mathcal{D}$  satisfies the top-two condition, there exists  $P'_i \in \mathcal{D}$  such that

$$x_i = r_1(P'_i, O) \quad \text{and} \quad o_i = r_2(P'_i, O).$$

Consider the profile  $(P'_S, P_{-S})$ . By individual rationality, it must be that either

1.  $\varphi_i(P'_S, P_{-S}) = x_i$  for all  $i \in S$ , or
2.  $\varphi_i(P'_S, P_{-S}) = o_i$  for all  $i \in S$ .

Were  $\varphi$  Pareto efficient, we would immediately have that  $\varphi$  executes the cycle. The following lemma shows that pair efficiency suffices under individual rationality and strategyproofness.

**Lemma 1.**  $\varphi_i(P'_S, P_{-S}) = x_i$  for all  $i \in S$ .

*Proof.* If  $|S| = 2$ , the claim follows immediately from pair efficiency, so assume  $|S| \geq 3$ . Label the agents  $S = \{i_1, i_2, \dots, i_{|S|}\}$  so that, for each  $k \in \{1, \dots, |S|\}$ , agent  $i_k$  most prefers the endowment of agent  $i_{k+1}$ , where we adopt the cyclic convention  $i_{|S|+1} = i_1$ .

| $P'_{i_1}$ | $P'_{i_2}$ | $\dots$  | $P'_{i_k}$    | $\dots$  | $P'_{i_{ S -1}}$ | $P'_{i_{ S }}$ |
|------------|------------|----------|---------------|----------|------------------|----------------|
| $o_{i_2}$  | $o_{i_3}$  | $\dots$  | $o_{i_{k+1}}$ | $\dots$  | $o_{i_{ S }}$    | $o_{i_1}$      |
| $o_{i_1}$  | $o_{i_2}$  | $\dots$  | $o_{i_k}$     | $\dots$  | $o_{i_{ S -1}}$  | $o_{i_{ S }}$  |
| $\vdots$   | $\vdots$   | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$         | $\vdots$       |

Table 2: Agents in cycle  $S = \{i_1, \dots, i_{|S|}\}$  rank their endowment second.

We construct a sequence of profiles differing only in  $i_1$ 's preference. For  $k \in \{2, 3, \dots, |S|\}$ , both  $o_{i_k}, o_{i_{k+1}} \in r_1(\mathcal{D}, O)$ , and since  $\mathcal{D}$  satisfies the top-two condition, there exists  $P_{i_1}^k \in \mathcal{D}$  such that

$$o_{i_k} = r_1(P_{i_1}^k, O) \quad \text{and} \quad o_{i_{k+1}} = r_2(P_{i_1}^k, O).$$

Define the profile  $P^k = (P_{i_1}^k, P'_{S \setminus \{i_1\}}, P_{-S})$ .

| $P_{i_1}^k$   | $P'_{i_2}$ | $\dots$  | $P'_{i_k}$    | $\dots$  | $P'_{i_{ S -1}}$ | $P'_{i_{ S }}$ |
|---------------|------------|----------|---------------|----------|------------------|----------------|
| $o_{i_k}$     | $o_{i_3}$  | $\dots$  | $o_{i_{k+1}}$ | $\dots$  | $o_{i_{ S }}$    | $o_{i_1}$      |
| $o_{i_{k+1}}$ | $o_{i_2}$  | $\dots$  | $o_{i_k}$     | $\dots$  | $o_{i_{ S -1}}$  | $o_{i_{ S }}$  |
| $\vdots$      | $\vdots$   | $\ddots$ | $\vdots$      | $\ddots$ | $\vdots$         | $\vdots$       |

Table 3: Agent  $i_1$ 's report used to move backward through the cycle using pair efficiency.

We show that for every  $k \in \{2, 3, \dots, |S|\}$ ,

$$\varphi_{i_1}(P^k) = o_{i_k}.$$

First consider  $k = |S|$ . By individual rationality and pair efficiency,  $\varphi_{i_1}(P^{|S|}) = o_{i_{|S|}}$ . Now consider any  $k \in \{2, \dots, |S| - 1\}$  and assume  $\varphi_{i_1}(P^{k+1}) = o_{i_{k+1}}$ . Since  $i_1$  can obtain  $o_{i_{k+1}}$  by reporting  $P_{i_1}^{k+1}$ , strategyproofness implies that  $\varphi_{i_1}(P^k) \in \{o_{i_k}, o_{i_{k+1}}\}$ . But if  $i_1$  gets  $o_{i_{k+1}}$ , by individual rationality,  $i_k$  should get  $o_{i_k}$ , which violates pair efficiency. Therefore,  $\varphi_{i_1}(P^k) = o_{i_k}$ .

Thus, for every  $k \in \{2, 3, \dots, |S|\}$ ,  $\varphi_{i_1}(P^k) = o_{i_k}$ . In particular,  $\varphi_{i_1}(P^2) = o_{i_2}$ . By strategyproofness,  $\varphi_{i_1}(P'_S, P_{-S}) = o_{i_2}$ . By individual rationality,  $\varphi_i(P'_S, P_{-S}) = x_i$  for all  $i \in S$ .  $\square$

Returning to the proof of the theorem, Lemma 1 gives  $\varphi_i(P'_S, P_{-S}) = x_i$  for all  $i \in S$ . We now return from  $P'_S$  to  $P_S$  one agent at a time. Fix any  $j \in S$ . By strategyproofness,  $\varphi_j(P_j, P'_{S \setminus \{j\}}, P_{-S}) = x_j$ , and by individual rationality, for each  $i \in S$ ,

$$\varphi_i(P_j, P'_{S \setminus \{j\}}, P_{-S}) = x_i.$$

Now suppose for any  $T \subset S$  where  $|T| \leq k < |S|$ , we have that for each  $i \in S$ ,

$$\varphi_i(P_T, P'_{S \setminus T}, P_{-S}) = x_i.$$

Fix any  $T$  of size  $k + 1$  and any  $j \in T$ . By strategyproofness,  $\varphi_j(P_T, P'_{S \setminus T}, P_{-S}) = x_j$ , and in fact, for any  $i \in T$ ,  $\varphi_i(P_T, P'_{S \setminus T}, P_{-S}) = x_i$ . Now there must be some  $i \in T$  such that  $x_i = o_r$  where  $r \in S \setminus T$ . By individual rationality, for each  $i \in S$ ,  $\varphi_i(P_T, P'_{S \setminus T}, P_{-S}) = x_i$ . It follows by induction that for each  $i \in S$ ,

$$\varphi_i(P_S, P_{-S}) = \varphi_i(P) = x_i.$$

Thus,  $\varphi$  must execute the cycle in the first round of TTC at  $P$ .

We can now apply the same argument iteratively to show that  $\varphi$  must also execute the subsequent cycles of the TTC algorithm. Note that the argument above does not depend on the reports of agents outside  $S$ , and hence, for every  $Q_{-S} \in \mathcal{D}^{N \setminus S}$ ,

$$\varphi_i(P_S, Q_{-S}) = x_i \quad \text{for all } i \in S.$$

In words,  $\varphi$  executes the cycle  $S$  regardless of the reports of the remaining agents. This allows us to remove the agents in  $S$  and their assigned objects, and apply the same argument to the reduced economy. The remaining objects form some subset  $O' \subset O$ . Since  $\mathcal{D}$  satisfies the top-two condition within  $O'$ , the argument applies with  $O'$  in place of  $O$ . Iterating over all TTC cycles yields  $\varphi(P) = x = TTC(P)$ . Since  $P \in \mathcal{D}^N$  was arbitrary,  $\varphi = TTC$ .  $\square$

We now compare our proof of Theorem 1 with existing proofs characterizing TTC on the unrestricted domain. To the best of our knowledge, all existing proofs, whether using Pareto efficiency (Ma (1994)), or pair efficiency (Ekici (2024)), construct auxiliary profiles of the form displayed in Table 2, where agents in a trading cycle rank their own endowment second. At such profiles, individual rationality and Pareto efficiency immediately imply that  $\varphi$  must execute the cycle. A key challenge is to establish this using pair efficiency.

While the existing proofs with pair efficiency (Ekici (2024), Ekici and Sethuraman (2024)) differ in their underlying approaches, we focus on how they overcome this specific challenge and reinterpret their arguments in our context. Ekici (2024) employs a contradiction argument. Assuming  $\varphi$  satisfies the axioms but differs from TTC, there is some profile at which  $\varphi$  does not execute a TTC cycle. Taking  $S = \{i_1, \dots, i_{|S|}\}$  to be a smallest such cycle across profiles, the argument proceeds by showing that, at the auxiliary profile in Table 2,  $\varphi$  must assign each agent in  $S$  their endowment. From here, if agent  $i_1$  instead reports preferences whose top-three objects are  $o_{i_2} o_{i_3} o_{i_1}$  and subsequently  $o_{i_3} o_{i_1} o_{i_2}$ ,  $\varphi$  must continue to assign  $i_1$  their endowment. But then, in the second case,  $\varphi$  fails to execute a smaller cycle of size  $|S| - 1$ , contradicting the minimality of  $S$ . In a shorter proof, Ekici and Sethuraman (2024) show that  $\varphi$  must execute the cycle  $S$  at a profile of the form in Table 2; otherwise, agent  $i_1$  can instead report a preference  $o_{i_2} o_{i_3} \dots o_{i_{|S|}} o_{i_1}$ , where individual rationality and pair efficiency imply that  $\varphi$  must assign  $o_{i_2}$  to  $i_1$ , thereby violating strategyproofness.

In comparison, our proof uses preferences of the form displayed in Table 3, where agent  $i_1$  reports  $o_{i_k}$  and  $o_{i_{k+1}}$  as their top two objects. These preferences require no additional richness of the domain beyond what is needed to construct the auxiliary profile in Table 2. Thus, our proof uses minimal richness relative to the existing arguments. In addition, its direct approach illuminates the role played by each axiom in forcing the rule to coincide with TTC. The existing proofs have their own advantages: Ekici (2024)'s proof offers a novel approach based on a notion of distance between mechanisms, and the proof of Ekici and Sethuraman (2024) does not exploit the details of the TTC rule, which may be valuable in other settings.

Theorem 1 enables us to classify several important domains as TTC domains. We first consider the single-dipped domain, which has already been shown to be a TTC domain by Tamura (2023), Hu and Zhang (2024).

**Example 1** (Single-dipped domain). *Suppose  $\mathcal{D}^{SD} \subset \mathcal{P}$  contains preferences single-dipped with respect to the ordering  $o_1 \rightarrow \dots \rightarrow o_n$ :*

$$\mathcal{D}^{SD} = \{P_0 \in \mathcal{P} : o_d = r_n(P_0, O) \implies o_k P_0 o_{k+1} \text{ for } k < d \text{ and } o_{k+1} P_0 o_k \text{ for } k \geq d\}.$$

*For any  $n \geq 2$ , any  $O' \subset O$  with  $|O'| \geq 2$ , only the two extreme objects of  $O'$  with respect to  $o_1 \rightarrow \dots \rightarrow o_n$  can be ranked first within  $O'$ , and they can be ranked as the top two in both possible orders. Thus,  $\mathcal{D}^{SD}$  satisfies the top-two condition within every  $O' \subset O$ , and by Theorem 1, is a TTC domain.*

We next consider the partial-agreement domain, which, to the best of our knowledge, has not previously been explored in this context.<sup>5</sup> This domain captures environments in which there is an objective dominance relation over objects. For example, in kidney exchange, younger donor kidneys may dominate older donor kidneys, making preferences that violate this dominance relation implausible.

**Example 2** (Partial agreement domain). *Suppose  $\mathcal{D}^{PA}(\succ) \subset \mathcal{P}$  contains preferences consistent with the partial order  $\succ$  over  $O$ :*

$$\mathcal{D}^{PA}(\succ) = \{P_0 \in \mathcal{P} : \text{for all } a, b \in O, a \succ b \implies a P_0 b\}.$$

*For any  $n \geq 2$ , any  $O' \subset O$  with  $|O'| \geq 2$ , only the undominated objects within  $O'$  can be ranked first, and they can be ranked as the top two in both possible orders. Thus, for any partial order  $\succ$  over  $O$ ,  $\mathcal{D}^{PA}(\succ)$  satisfies the top-two condition within every  $O' \subset O$ , and by Theorem 1, is a TTC domain. Note the two extreme cases: an empty partial order corresponds to the unrestricted domain  $\mathcal{D} = \mathcal{P}$ , while a complete order yields a domain containing a single preference, where individual rationality alone characterizes the TTC rule.*

We conclude this section with a remark illustrating how our analysis can be used to derive analogous results for settings in which domain restrictions are heterogeneous across agents. In this more general framework, we also provide an example in which Ma (1994)'s characterization holds, but that of Ekici (2024) does not.

**Remark 1** (Heterogeneous domain restrictions). *Suppose that each agent  $i \in N$  has a preference domain  $\mathcal{D}_i \subset \mathcal{P}$ . If, for every  $i \in N$ ,  $\mathcal{D}_i$  satisfies the top-two condition and  $o_i \in r_1(\mathcal{D}_i, O)$ , then TTC is the only rule  $\varphi : \mathcal{D}_1 \times \cdots \times \mathcal{D}_n \rightarrow \mathcal{X}$  that is individually rational, Pareto efficient, and strategyproof.<sup>6</sup> Indeed, these assumptions ensure that the auxiliary profile in Table 2 can always be constructed, so the same argument using Pareto efficiency in the proof of Theorem 1 forces  $\varphi$  to coincide with TTC.*

*Under these conditions, however, Pareto efficiency cannot be replaced by pair efficiency, because preferences in Table 3 may not be available. To see this, let  $n = 3$  and consider*

$$\mathcal{D}_1 = \{o_1 o_2 o_3, o_2 o_1 o_3\}, \mathcal{D}_2 = \{o_2 o_3 o_1, o_3 o_2 o_1\}, \text{ and } \mathcal{D}_3 = \{o_3 o_1 o_2, o_1 o_3 o_2\}.$$

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<sup>5</sup>Nicolo and Rodriguez-Alvarez (2017), Fujinaka and Wakayama (2024) study a related domain, which they refer to as the common-ranking domain.

<sup>6</sup>We thank the Associate Editor for this suggestion.

Notice that for every  $i$ ,  $\mathcal{D}_i$  satisfies the top-two condition and contains a preference that ranks  $o_i$  first. From above, Ma (1994)'s characterization holds. Nevertheless, the rule that always selects the endowment allocation is individually rational, pair efficient, and strategyproof. Moreover, at the profile  $P_1 = o_2o_1o_3$ ,  $P_2 = o_3o_2o_1$ , and  $P_3 = o_1o_3o_2$ , this rule differs from TTC. Thus, the analogous characterization of Ekici (2024) does not hold.

If the assumptions on the domains were strengthened to ensure that the auxiliary profiles in both Tables 2 and 3 could be constructed, then the characterization using pair efficiency would follow as well.

### 3.3 Is top-two necessary?

We now turn to whether the top-two condition is also necessary for a domain to be a TTC domain. Equivalently, if a domain  $\mathcal{D} \subset \mathcal{P}$  fails the top-two condition, must there exist a rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$ , distinct from TTC, that satisfies the axioms in Theorem 1? A natural approach is to construct such a rule directly. The difficulty is that the failure of the top-two condition provides only limited information about the domain: there exists a subset of objects within which two objects can each be ranked first, but cannot be ranked as the top two.

Nevertheless, our proof of Theorem 1 provides useful guidance for approaching the construction. Specifically, it establishes the following stronger claim: on any domain  $\mathcal{D} \subset \mathcal{P}$  and at any profile  $P \in \mathcal{D}^N$ , any rule  $\varphi$  satisfying the axioms must execute the TTC trading cycles as long as any two objects in the relevant cycle can be ranked as the top two objects within the set of remaining objects at that stage. In particular, even if  $\mathcal{D}$  does not satisfy the top-two condition, the proof still implies  $\varphi(P) = \text{TTC}(P)$  whenever this condition holds successively along the reduced object sets generated by TTC at  $P$ . Thus,  $\varphi$  can deviate from TTC only when the failure of the top-two condition prevents the construction of the auxiliary preferences displayed in Tables 2 and 3, which, as shown in the proof of Theorem 1, force  $\varphi$  to execute the TTC trading cycle.

We use this observation to obtain a non-TTC rule satisfying the axioms for a broad subclass of domains that fail the top-two condition. Specifically, we construct such a rule for domains in which there exists a small subset  $O' \subset O$ , with  $|O'| \leq 4$ , such that the domain fails the top-two condition within  $O'$ , and every object outside  $O'$  can be ranked first within  $O' \cup \{o\}$ . The following result formalizes this partial converse.

**Theorem 2.** *Suppose  $\mathcal{D} \subset \mathcal{P}$  does not satisfy the top-two condition. Suppose further that there exists a subset  $O' \subset O$  such that:*

- (i)  $\mathcal{D}$  does not satisfy the top-two condition within  $O'$ ;*
- (ii)  $|O'| \leq 4$ ;*
- (iii) for every  $o \in O \setminus O'$ ,  $o \in r_1(\mathcal{D}, O' \cup \{o\})$ .*

*Then there exists a rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$ , distinct from TTC, that is individually rational, Pareto efficient, and strategyproof.*

The proof proceeds in two steps. First, we consider the special case in which the top-two condition fails on the full object set  $O$  and  $|O| \leq 4$  (Lemma 2). Using the fact that there exist profiles where an agent in a TTC trading cycle cannot report their endowment as their second-ranked object, we construct a rule that deviates from TTC on a carefully chosen nonempty subset of such profiles by assigning this agent their second-ranked object instead of their TTC assignment. The choice of this subset and the specific deviation from TTC ensure that the rule satisfies the axioms in the theorem.

In the second step, we use this baseline construction to construct a rule for economies with an arbitrary number of objects, provided there is a subset  $O' \subset O$  satisfying conditions (i)–(iii). To illustrate, suppose  $O' = \{o_1, o_2, o_3, o_4\}$ . Our rule coincides with TTC unless every agent  $i \in \{5, \dots, n\}$  ranks their endowment  $o_i$  first within  $O' \cup \{o_i\}$ , in which case our rule separates the economy into two parts: agents  $\{5, \dots, n\}$  receive their TTC assignments, while agents  $\{1, \dots, 4\}$  are assigned according to the rule constructed in the first step. Since the first-step rule differs from TTC on some profiles and satisfies the axioms, this construction yields a rule distinct from TTC that satisfies the axioms in the theorem. We now present the full proof.

*Proof.* The proof is constructive. We first prove the result for the special case in which the top-two condition fails within the full object set and there are at most four objects. We then extend the construction to economies with an arbitrary number of objects, provided there exists a subset  $O' \subset O$  satisfying conditions (i) – (iii) of the theorem.

**Lemma 2.** *Suppose  $n \leq 4$  and  $\mathcal{D}$  does not satisfy the top-two condition within  $O$ . Then there exists a rule  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$ , distinct from TTC, that is individually rational, Pareto efficient, and strategyproof.*

*Proof.* Since  $\mathcal{D}$  does not satisfy the top-two condition within  $O$ , there exist two objects that can each be ranked first, but cannot be ranked as the top two in some order. Relabeling objects if necessary, assume  $\mathcal{D}$  is such that:

1.  $o_1, o_2 \in r_1(\mathcal{D}, O)$ ,
2. for all  $P_0 \in \mathcal{D}$ ,  $o_2 = r_1(P_0, O) \implies o_1 \neq r_2(P_0, O)$ ,
3. there exists  $P_0 \in \mathcal{D}$  such that  $o_2 P_0 o_3 P_0 \cdots P_0 o_n$ .

Define the following set of preference profiles:

$$\text{Diff} = \{P \in \mathcal{D}^N : r_1(P_1, O) = o_2, \text{ and for every } i \geq 2, r_1(P_i, \{o_{i-1}, o_i, \dots, o_n\}) = o_{i-1}\}.$$

Define  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  as follows:

$$\varphi(P) = \begin{cases} \text{TTC}(P) & \text{if } P \notin \text{Diff}, \\ (o_k, o_1, o_2, \dots, o_{k-1}, \text{TTC}(P_{\{k+1, \dots, n\}} | \{o_{k+1}, \dots, o_n\})) & \text{if } P \in \text{Diff and } o_k = r_2(P_1, O). \end{cases}$$

In words,  $\varphi$  coincides with TTC outside Diff. On profiles in Diff, which is nonempty by the assumptions above,  $\varphi$  assigns agent 1 their second-ranked object  $o_k$  instead of their TTC assignment  $o_2$ . This deviation is sustainable because  $\mathcal{D}$  does not allow agent 1 to rank their endowment  $o_1$  second below  $o_2$ . Hence, the auxiliary preference profile used in the proof of Theorem 1 cannot be constructed, and the argument does not force the execution of this TTC cycle. It is straightforward to verify that  $\varphi$  is individually rational and Pareto efficient. We now show that  $\varphi$  is strategyproof.

Fix an agent  $i \in N$  and fix  $P_{-i} \in \mathcal{D}^{N \setminus \{i\}}$ . Since Diff is defined by separate restrictions on each agent's preference, there are two possibilities. First,  $P_{-i}$  may be such that no report of agent  $i$  can make the profile belong to Diff:  $(P'_i, P_{-i}) \notin \text{Diff}$  for every  $P'_i \in \mathcal{D}$ . In this case,  $\varphi$  coincides with TTC for every report of agent  $i$ , and hence agent  $i$  cannot gain by misreporting. Second,  $P_{-i}$  may be such that agent  $i$ 's report determines whether the profile belongs to Diff: there exists  $P'_i, P''_i \in \mathcal{D}$  such that  $(P'_i, P_{-i}) \in \text{Diff}$  and  $(P''_i, P_{-i}) \notin \text{Diff}$ . We analyze this case separately for each agent.

1.  $i = 1$ : By definition of Diff,  $P_{-1}$  is such that for every  $i \geq 2$ ,  $r_1(P_i, \{o_{i-1}, o_i, \dots, o_n\}) = o_{i-1}$ . We claim that, for every  $P_1 \in \mathcal{D}$ ,

$$\varphi_1(P_1, P_{-1}) = r_1(P_1, O \setminus \{o_2\}).$$



If  $(P_1, P_{-1}) \in \text{Diff}$ , then  $r_1(P_1, O) = o_2$ , and by definition of  $\varphi$  agent 1 receives their second-ranked object in  $O$ , which is exactly  $r_1(P_1, O \setminus \{o_2\})$ .

If  $(P_1, P_{-1}) \notin \text{Diff}$ , then  $r_1(P_1, O) \neq o_2$ , and  $\varphi$  coincides with TTC under which agent 1 receives their first-ranked object in  $O$ , which is also  $r_1(P_1, O \setminus \{o_2\})$ .

It follows that agent 1 cannot gain by misreporting.

2.  $i = 2$ : For all  $P \in \mathcal{D}^N$ ,  $\varphi_2(P) = \text{TTC}_2(P)$ . Thus, agent 2 cannot gain by misreporting.
3.  $i = 3$ : By definition of Diff,  $P_{-3}$  is such that  $r_1(P_1, O) = o_2$  and, for every  $i \geq 2$  with  $i \neq 3$ ,  $r_1(P_i, \{o_{i-1}, o_i, \dots, o_n\}) = o_{i-1}$ . We claim that, for every  $P_3 \in \mathcal{D}$ ,

$$\varphi_3(P_3, P_{-3}) = r_1(P_3, O \setminus \{o_1\}).$$

If  $(P_3, P_{-3}) \in \text{Diff}$ , then  $r_1(P_3, O \setminus \{o_1\}) = o_2$ , and by definition of  $\varphi$  agent 3 receives  $o_2$ .

If  $(P_3, P_{-3}) \notin \text{Diff}$ , then  $r_1(P_3, O \setminus \{o_1\}) \neq o_2$ , and  $\varphi$  coincides with TTC under which agent 3 receives their first-ranked object in  $O \setminus \{o_1, o_2\}$ , which is also  $r_1(P_3, O \setminus \{o_1\})$ .

It follows that agent 3 cannot gain by misreporting.

4.  $i = 4$ : By definition of Diff,  $P_{-4}$  is such that  $r_1(P_1, O) = o_2$  and, for every  $i \geq 2$  with  $i \neq 4$ ,  $r_1(P_i, \{o_{i-1}, o_i, \dots, o_n\}) = o_{i-1}$ .

If  $(P_4, P_{-4}) \in \text{Diff}$ , then  $o_3 P_4 o_4$ , and by definition of  $\varphi$ , agent 4 receives either  $o_3$  or  $o_4$  depending on  $r_2(P_1, O)$ .

If  $(P_4, P_{-4}) \notin \text{Diff}$ , then  $o_4 P_4 o_3$ , and  $\varphi_4(P_4, P_{-4}) = o_4$ .

Thus, when  $o_3 P_4 o_4$ , agent 4 receives one of  $\{o_3, o_4\}$ , and when  $o_4 P_4 o_3$ , agent 4 receives  $o_4$ . Therefore, agent 4 cannot gain by misreporting.

Thus,  $\varphi$  is strategyproof.<sup>7</sup> □

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<sup>7</sup>The rule  $\varphi$  is well-defined for any  $n$  and any domain  $\mathcal{D} \subset \mathcal{P}$  satisfying the three conditions at the beginning of the proof of Lemma 2, but it need not be strategyproof for  $n > 4$ . To see the issue, suppose  $n = 5$  and consider a profile  $P \in \mathcal{D}^N$  at which TTC first forms a cycle between agents 1 and 2, then a cycle between agents 3 and 5, while agent 4 receives  $o_4$ . Suppose  $o_5 P_4 o_3 P_4 o_4$ , which implies  $P \notin \text{Diff}$  and hence  $\varphi(P) = \text{TTC}(P)$ . If agent 4 instead reports some  $P'_4 \in \mathcal{D}$  such that  $o_3 P'_4 o_5 P'_4 o_4$  and  $(P'_4, P_{-4}) \in \text{Diff}$ , it obtains  $o_3$  or  $o_5$  with this misreport, both of which are strictly preferred to  $o_4$  under  $P_4$ .

We now use the construction in Lemma 2 to prove the result. Suppose  $\mathcal{D}$  is such that there exists  $O' \subset O$  satisfying conditions (i)–(iii). Relabeling objects if necessary, let  $O' = \{o_1, o_2, \dots, o_k\}$ , where  $k \leq 4$ . For each  $i \in \{k+1, \dots, n\}$ , define

$$\mathcal{D}_i = \{P_0 \in \mathcal{D} : o_i = r_1(P_0, O' \cup \{o_i\})\}.$$

By condition (iii), each  $\mathcal{D}_i$  is nonempty. From Lemma 2, let  $\varphi'$  be a rule for the restricted economy on  $O'$ , distinct from TTC, that is individually rational, Pareto efficient, and strategyproof.

Define  $\varphi : \mathcal{D}^N \rightarrow \mathcal{X}$  as follows:

$$\varphi(P) = \begin{cases} \text{TTC}(P) & \text{if } P_i \notin \mathcal{D}_i \text{ for some } i \in \{k+1, \dots, n\}, \\ (\varphi'(P_{\{1, \dots, k\}}|O'), \text{TTC}(P_{\{k+1, \dots, n\}}|O \setminus O')) & \text{if } P_i \in \mathcal{D}_i \text{ for all } i \in \{k+1, \dots, n\}. \end{cases}$$

In words,  $\varphi$  coincides with TTC unless every agent  $i \in \{k+1, \dots, n\}$  reports  $P_i \in \mathcal{D}_i$ . On such profiles, which exist by condition (iii), the rule separates the economy into two parts. Agents in  $\{1, \dots, k\}$  are assigned objects in  $O'$  according to  $\varphi'$  from Lemma 2, while agents in  $\{k+1, \dots, n\}$  are assigned objects in  $O \setminus O'$  according to TTC. Since  $\varphi'$  differs from TTC, so does  $\varphi$ . It is straightforward to verify that  $\varphi$  is individually rational and Pareto efficient. We now show that  $\varphi$  is strategyproof.

For any agent  $i \in \{k+1, \dots, n\}$ , the definition of  $\mathcal{D}_i$  implies that for all  $P \in \mathcal{D}^N$ ,  $\varphi_i(P) = \text{TTC}_i(P)$ . This is because in the second branch, all agents in  $\{k+1, \dots, n\}$  rank their own endowment above every object in  $O'$ . Consequently, their TTC assignment in the restricted economy actually coincides with their TTC assignment under the full economy. Thus, these agents cannot gain by misreporting.

For agents in  $\{1, \dots, k\}$ , the branch of  $\varphi$  is determined entirely by the reports of agents in  $\{k+1, \dots, n\}$ , and hence cannot be affected by their own reports. Conditional on the branch, their assignment is determined either by TTC or by  $\varphi'$ , both of which are strategyproof. Hence, these agents also cannot gain by misreporting. Thus,  $\varphi$  is strategyproof.  $\square$

We note that Theorem 2 implies that the top-two condition is, in fact, necessary when  $n \leq 4$ . Indeed, when  $n = 3$  and  $\mathcal{D}$  fails the top-two condition, then the failure must occur within  $O$ , so Theorem 2 applies directly. When  $n = 4$ , let  $O' \subset O$  be a maximal subset

within which the top-two condition fails. If  $|O'| = 4$ , the theorem again applies directly. If  $|O'| = 3$ , maximality implies that every  $o \in O \setminus O'$  can be ranked first within  $O' \cup \{o\}$ ; otherwise, the failure would extend to  $O' \cup \{o\}$ . Thus, Theorem 2 applies. Together with the sufficiency result in Theorem 1, this yields the following characterization of TTC domains for  $n \leq 4$ .

**Corollary 1.** *Suppose  $n \leq 4$  and  $\mathcal{D} \subset \mathcal{P}$ . The following are equivalent:*

- (i)  $\mathcal{D}$  satisfies the top-two condition.
- (ii) There is a unique individually rational, Pareto efficient, and strategyproof rule on  $\mathcal{D}$ .
- (iii) There is a unique individually rational, pair efficient, and strategyproof rule on  $\mathcal{D}$ .

For  $n \geq 5$ , although Theorem 2 does not establish necessity of the top-two condition in full generality, it covers some important domains, thereby implying that TTC is not uniquely characterized by the axioms on those domains. Two such domains are the single-peaked domain (Example 3), for which Bade (2019) already proposed the crawler rule, and the circular domain (Example 4), which has been studied in social choice (Sato (2010)) but remains unexplored in this context.

**Example 3** (Single-peaked domain). *Suppose  $\mathcal{D}^{SP} \subset \mathcal{P}$  contains preferences single-peaked with respect to the ordering  $o_1 \rightarrow \dots \rightarrow o_n$ :*

$$\mathcal{D}^{SP} = \{P_0 \in \mathcal{P} : o_p = r_1(P_0, O) \implies o_{k+1} P_0 o_k \text{ for } k < p \text{ and } o_k P_0 o_{k+1} \text{ for } k \geq p\}.$$

*For any  $n \geq 3$ , consider  $O' = \{o_1, o_2, o_3\}$ . While  $o_1, o_3 \in r_1(\mathcal{D}^{SP}, O')$ , there is no  $P_0 \in \mathcal{D}^{SP}$  such that  $o_1 = r_1(P_0, O')$ ,  $o_3 = r_2(P_0, O')$ , so  $\mathcal{D}^{SP}$  does not satisfy the top-two condition within  $O'$ . Moreover, every  $o \in \{o_4, \dots, o_n\}$  can be ranked first within  $\{o_1, o_2, o_3, o\}$ . Thus,  $O'$  satisfies all three conditions of Theorem 2, and it follows that  $\mathcal{D}^{SP}$  is not a TTC domain.*

**Example 4** (Circular domain). *Suppose  $\mathcal{D}^C \subset \mathcal{P}$  contains preferences described by a first-ranked object, and a clockwise or counterclockwise traversal along the cyclic order  $o_1 \rightarrow o_2 \rightarrow \dots \rightarrow o_n \rightarrow o_1$ :*

$$\mathcal{D}^C = \{P_0 \in \mathcal{P} : o_p = r_1(P_0, O) \implies P_0 \in \{o_p \dots o_n o_1 \dots o_{p-1}, o_p \dots o_1 o_n \dots o_{p+1}\}\}.$$

*For any  $n \geq 4$ , consider  $O' = \{o_1, o_2, o_3, o_4\}$ . While  $o_1, o_3 \in r_1(\mathcal{D}^C, O')$ , there is no  $P_0 \in \mathcal{D}^C$  such that  $o_1 = r_1(P_0, O')$ ,  $o_3 = r_2(P_0, O')$ , so  $\mathcal{D}^C$  does not satisfy the top-two condition within  $O'$ . Moreover, every  $o \in \{o_5, \dots, o_n\}$  can be ranked first within  $\{o_1, o_2, o_3, o_4, o\}$ . Thus,  $O'$  satisfies all three conditions of Theorem 2, and it follows that  $\mathcal{D}^C$  is not a TTC domain.*

As discussed in the above examples, both the single-peaked and circular domains fail the top-two condition and satisfy the conditions of Theorem 2. Consequently, the proof of Theorem 2 constructs a rule distinct from TTC that satisfies individual rationality, Pareto efficiency, and strategyproofness on each domain. Although the resulting rule may not be of independent interest for any particular domain, the construction serves to demonstrate uniformly that a non-TTC rule satisfying the axioms exists for every domain covered by Theorem 2.

Lastly, we present an example of a domain that fails the top-two condition but is not covered by Theorem 2 when  $n \geq 5$ . The *one-pair exclusion domain* contains all preferences over  $O$  except those that rank  $o_i$  first and  $o_j$  second. As shown in Example 5, this domain fails the top-two condition, and does so only on the full set  $O$ .

**Example 5** (One-pair exclusion domain). *Suppose  $\mathcal{D}^{-o_i o_j \dots} \subset \mathcal{P}$  contains all preferences over  $O$  except those that rank  $o_i$  first and  $o_j$  second:*

$$\mathcal{D}^{-o_i o_j \dots} = \mathcal{P} \setminus \{P_0 \in \mathcal{P} : r_1(P_0, O) = o_i, r_2(P_0, O) = o_j\}.$$

*For any  $n \geq 3$ ,  $\mathcal{D}^{-o_i o_j \dots}$  fails the top-two condition within  $O$ : both  $o_i$  and  $o_j$  can be ranked first, but there is no  $P_0 \in \mathcal{D}^{-o_i o_j \dots}$  that ranks  $o_i$  first and  $o_j$  second. At the same time, for every proper subset  $O' \subsetneq O$ , the domain admits arbitrary rankings of objects within  $O'$ , and hence, satisfies the top-two condition within  $O'$ . Thus,  $\mathcal{D}^{-o_i o_j \dots}$  fails the top-two condition only on the full set  $O$ . In particular, for  $n \geq 5$ , there is no subset  $O' \subset O$  satisfying conditions (i)-(iii) of Theorem 2, and so this domain is not covered by the result.*

In addition to clarifying the scope of Theorem 2, the one-pair exclusion domain is especially useful because, up to relabeling of objects, it represents a maximal domain failing the top-two condition. To see this, consider any domain that fails the top-two condition. Then there exist a subset  $O' \subset O$  and two objects  $o_i, o_j \in O'$  such that no preference in the domain ranks  $o_i$  first and  $o_j$  second within  $O'$ . It follows that the domain must exclude all preferences that rank  $o_i$  first and  $o_j$  second on  $O$ ; otherwise, restricting such a preference to  $O'$  would rank  $o_i$  first and  $o_j$  second within  $O'$ . Therefore, up to relabeling of objects, every domain failing the top-two condition is contained in a one-pair exclusion domain. Since the one-pair exclusion domain contains  $n! - (n - 2)!$  preferences, any domain with more than  $n! - (n - 2)!$  preferences must satisfy the top-two condition, and hence is a TTC domain by Theorem 1. Importantly, a construction for the one-pair exclusion domain can be applied

analogously to any domain failing the top-two condition, with the deviation anchored at a suitable profile in the relevant domain. Therefore, constructing a rule distinct from TTC that satisfies the axioms on the one-pair exclusion domain would establish necessity of the top-two condition in general.

## 4 Conclusion

We introduce the top-two condition on preference domains and establish it as a valuable tool for analyzing object reallocation problems under domain restrictions. The condition requires that within every subset of objects, if two objects can each be ranked first, then they can also be ranked as the top two in both possible orders. We show that standard characterizations of TTC on the unrestricted domain extend to every restricted domain that satisfies the condition. As a partial converse, for domains that fail the condition within a small subset of objects and satisfy a mild extension condition, we construct a non-TTC rule satisfying individual rationality, Pareto efficiency, and strategyproofness. An important question for future research is to better understand the structure of rules satisfying these axioms on domains that fail the top-two condition. In particular, it would be useful to know whether such rules can execute the longer trading cycles that TTC precludes at profiles such as the one in Table 1. Further work could also explore analogous conditions for other characterizations of TTC.

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